

Software Workflow and Design Methodology

Objective

The purpose of this tool is to determine whether the reinforcement percentage (pt) selected for dowel bars in a horizontal precast wall connection is adequate under all ETABS load combinations.

The program does not directly design dowel bars. Instead, it converts the specified reinforcement percentage into an equivalent reinforced concrete wall section and evaluates its axial-moment capacity using a P-M interaction curve generated from IS 456 stress block principles.

The selected reinforcement percentage is considered acceptable only if all load combinations satisfy the interaction requirements.

Overall Design Workflow

The complete workflow consists of eight sequential stages.

Stage 1 – ETABS Force Extraction

The user selects:

- Wall Label
- Floor Level

The program reads all corresponding ETABS load combinations and extracts:

- Axial force (P)
- Minor axis moment (M2)
- Major axis moment (M3)

The extracted values are copied into the Interaction Values worksheet.

Sign Convention

The software uses absolute values:

$$P = |P| \quad M2 = |M2| \quad M3 = |M3|$$

Therefore:

- Compression and tension are not distinguished.
- Positive and negative moments are not distinguished.

- Only force magnitudes are evaluated.
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Stage 2 – Calculation of Design Eccentricities

Accidental Eccentricity

As per IS 456 minimum eccentricity provisions:

$$eacc_major = H/500 + D/30$$

$$eacc_minor = H/500 + b/30$$

where:

H = effective height

D = wall depth

b = wall length

Total Eccentricity

The user-defined external eccentricities are added:

$$emajor = eext_major + eacc_major$$

$$eminor = eext_minor + eacc_minor$$

Stage 3 – Determination of Design Moments

For each load combination:

Minimum Design Moment

$$Mmin_major = P \times eacc_major$$

$$Mmin_minor = P \times eacc_minor$$

Additional Moment due to User Eccentricity

$$Mext_major = P \times eext_major$$

$$Mext_minor = P \times eext_minor$$

Final Design Moments

$$\mu_{major} = \max(M_3 + M_{ext_major}, M_{min_major})$$

$$\mu_{minor} = \max(M_2 + M_{ext_minor}, M_{min_minor})$$

These moments are used in subsequent interaction checks.

Stage 4 – Conversion of Reinforcement Percentage into Bar Layout

The user inputs:

- Reinforcement percentage (pt)
- Number of bars along depth
- Number of bars along width

The software first calculates:

$$A_{st} = pt \times b \times D / 100$$

Total number of bars:

$$N_{bars} = 2(N_{major} + N_{minor}) - 4$$

The equivalent main bar diameter is then calculated as:

$$\phi = \sqrt{(4A_{st} / \pi N_{bars})}$$

Therefore:

The interaction analysis is performed using an equivalent reinforcement arrangement derived from pt rather than directly using dowel bar diameter.

Stage 5 – Generation of P-M Interaction Curve

The software generates a complete axial load-moment interaction curve.

Neutral Axis Iteration

The neutral axis depth is varied from:

$$x_u = -0.01D$$

to

$$x_u = 10D$$

using 199 equally spaced increments.

For each neutral axis position:

1. Concrete compression force is calculated.
2. Concrete moment is calculated.
3. Steel strains are calculated using strain compatibility.
4. Steel stresses are obtained through interpolation of the IS 456 design stress-strain curve.
5. Steel forces and moments are calculated.
6. Total axial force and moment capacities are determined.

The resulting normalized capacities are stored as:

$$P_u/(f_{ck} \cdot b \cdot D)$$

$$M_u/(f_{ck} \cdot b \cdot D^2)$$

These values form the interaction curve.

Stage 6 – Capacity Determination by Interpolation

For every ETABS load combination:

The required normalized axial force ratio:

$$P_u/(f_{ck} \cdot b \cdot D)$$

is calculated.

The software then identifies the two adjacent points on the interaction curve.

Linear interpolation is performed:

$$y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$$

to determine the corresponding normalized moment capacity.

This provides the exact moment capacity associated with the required axial load.

Stage 7 – Major and Minor Axis Evaluation

The procedure is executed twice:

Major Axis Evaluation

Wall dimensions are considered in their original orientation.

Moment demand:

M_{u_major}

Capacity:

$M_{u_major_capacity}$

Minor Axis Evaluation

Wall dimensions are interchanged.

Moment demand:

M_{u_minor}

Capacity:

$M_{u_minor_capacity}$

Two independent interaction curves are therefore generated.

Limitation

The program performs two separate uniaxial interaction checks.

A full biaxial interaction surface is not generated.

Stage 8 – Interaction Check and Safety Assessment

For every load combination:

An interaction ratio is calculated in worksheet formulas.

The governing criterion is:

Interaction Ratio ≤ 1.0

If:

Interaction Ratio ≤ 1.0

Status = SAFE

Else:

Status = UNSAFE

Governing Load Combination Selection

After evaluating all load combinations:

The software identifies:

Maximum Interaction Ratio

This load combination is considered the governing case.

The governing values are automatically transferred to the final calculation sheet and report.

Important:

The governing load combination is selected based on maximum interaction ratio, not maximum axial load or maximum moment.

Scope of Design Check

The software explicitly checks:

✓ Axial compression capacity

✓ Moment capacity

✓ P-M interaction compliance

✓ Minimum eccentricity requirements

✓ Major-axis bending

✓ Minor-axis bending

✓ Governing load combination identification

Items Outside the Scope of the Tool

The following checks are not explicitly performed within this software:

✗ Development length

✗ Anchorage length

✗ Dowel pull-out resistance

✗ Dowel shear capacity

✗ Sleeve capacity

✗ Grout crushing

✗ Interface shear transfer

✗ Shear friction

✗ Local bearing around sleeves

✗ Construction tolerances

✗ Biaxial interaction analysis

✗ Slenderness moment magnification

These checks must be verified separately if required by the project specifications.

Fundamental Assumptions

1. Plane sections remain plane.
2. IS 456 stress block is applicable.
3. Reinforcement is symmetrically distributed.

4. Loads imported from ETABS are factored loads.
5. Compression force is treated as positive.
6. Moment sign is ignored and only magnitudes are evaluated.
7. Reinforcement percentage adequately represents the dowel reinforcement system.
8. Perfect bond exists between steel, grout and surrounding concrete.
9. Sectional strength governs the design of the horizontal wall connection.